

Control Systems (EEE/ECE/INSTR F242)

Tutorial 4 (3 February 2026)

1. Figure 1 shows an armature-controlled DC motor. Find the transfer function $\theta(s)/E_a(s)$ for this motor.

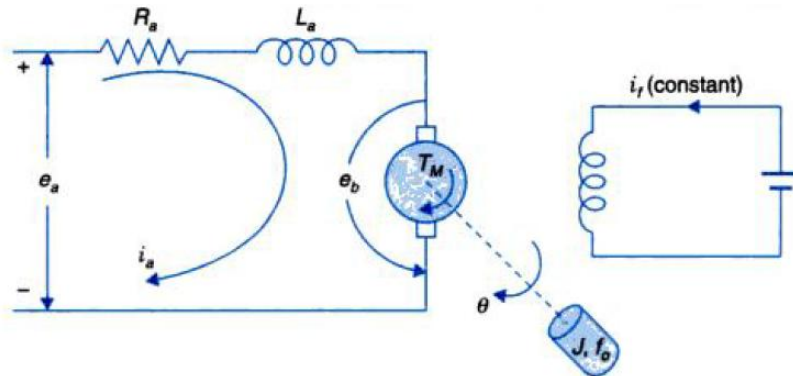


Fig. 1

2. The transfer function of an armature-controlled DC motor is employed to control the speed of J, f load in closed-loop with tachometer feedback as shown in Fig. 2. The motor transfer function is, $G(s) = K_m/(\tau's + 1)$.
- Determine the value of K such that time constant for closed loop system ($T(s) = \omega_o(s)/V_r(s)$), (say τ'') is $0.1\tau'$. Given, $\tau' = 0.1$, $K_m = 1.25$, $K_t = 0.2$.
 - Find expression for sensitivity $S_{K_t}^T$ and its value for $\omega = 0$ rad/sec.

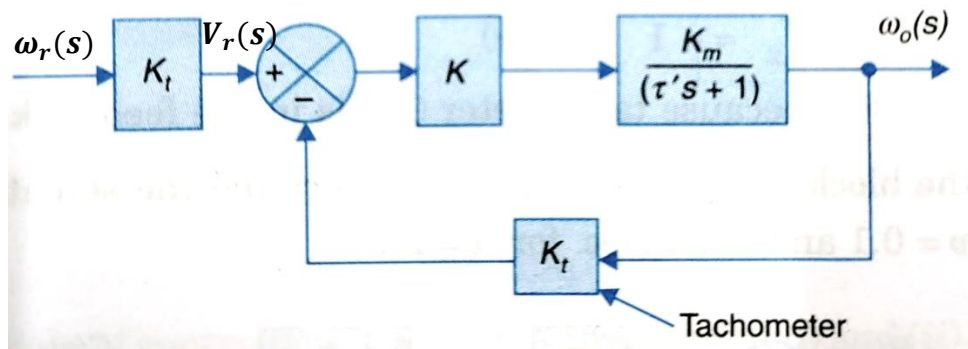


Fig. 2

3. Find the transfer function, $V_L(s)/V(s)$, for the electrical network shown in Fig. 3, which contains a nonlinear resistor whose voltage-current relationship is defined by $i_r = 2e^{0.1V_r}$, where i_r and V_r are the resistor current and voltage, respectively.

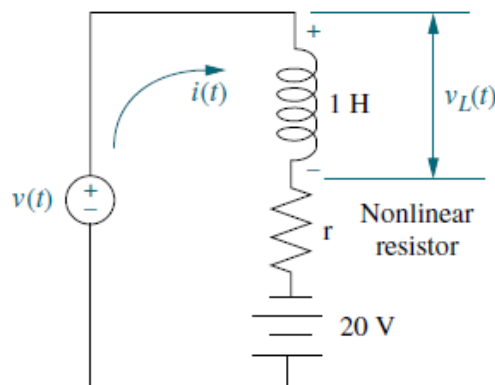


Fig. 3
